

IC-34: ELECTRIC FIELD AND EQUI-POTENTIALS

Rev 4-23-2023

FOR IN-CLASS ONLY

34.1 OBJECTIVE

To draw the electric field and equi-potential lines surrounding two or more electric charges, between charged parallel plates, and for other geometries.

34.2 EQUIPMENT

Field Mapping Board	Conducting Sheets with grid
Conducting paint	Connecting wires
Multimeter	Graph papers with same patterns as conducting sheets.
Conducting push pins	Power Supply



34.3 THEORY

Every charge has an electric field surrounding it. The strength of the electric field at any location is the magnitude of the electric force on a unit positive charge placed at that location. Force is a vector, and so is the electric field, the direction of the electric field being the same as the direction of the force. To determine the electric field, we could place a unit charge at different locations and measure the force that it experiences. The magnitude and directions of the electric field could then be shown by arrows to create a map of the electric field. The length of the arrows would represent the magnitude of the field. The arrows can also be joined by smooth lines to create a map showing the lines of force, or the lines of electric field. Then the closer the lines are, the stronger would be the field.

If the source charge (the charge for which we want to map the electric field) is positive, the force on the unit positive charge will be pointed radially away from the source charge (Figure 1). If the source charge is negative, the electric field will point radially towards it. We will mark the directions on the electric field lines. If there are multiple charges, each charge will have its own field lines which would tend to intersect each other. However, since the unit charge can only experience a force in one direction, there cannot be two lines of force

at any point in space. The electric field lines then bend so that they never cross each other (Figure 2). It is this pattern due to multiple charges that we will draw in this lab.

The equipotential lines are the lines surrounding a charge that are at the same potential (or voltage). For a single point charge in space, a spherical surface centered at the charge will be an equipotential surface – all points on it being at the same potential. Again, if there are multiple charges in a region, the equipotential surfaces bend and merge together, but do not intersect. Since $E = -dV/dr$, the equipotential surfaces are perpendicular to the electric field lines at every point (Figure 3).

In this experiment, we will stay in a two-dimensional plane, and so we will get electric field lines and equipotential lines on this flat 2-D surface. Since electric current flows in the direction of reducing electric field, we will find the directions of the electric field line by finding the direction in which the current flows. This in turn, is done by finding the direction in which there is a maximum potential drop.

The equipotential lines are found by placing the two prongs of a galvanometer at two close points. If there is no current, that means that the two points are at the same potential. Alternately, we could use a voltmeter to measure the potential difference between the two points. A zero potential difference would mean that the two points are on an equipotential line.

To create the 2-D surface, we will use a carbonized paper. Due to the slight electrical conductivity of carbon, this becomes a conducting paper (sometimes also called a resistance paper). A Silver paint is used to draw the electrodes of different shapes on the Conducting paper. The painted regions then become the electrodes, which are connected to a power source (or battery). A small electric current then flows from the positive to the negative electrodes.

The pattern of Electric Field Lines and Equipotential Lines for different charge configurations are shown in figures 1, 2 and 3.

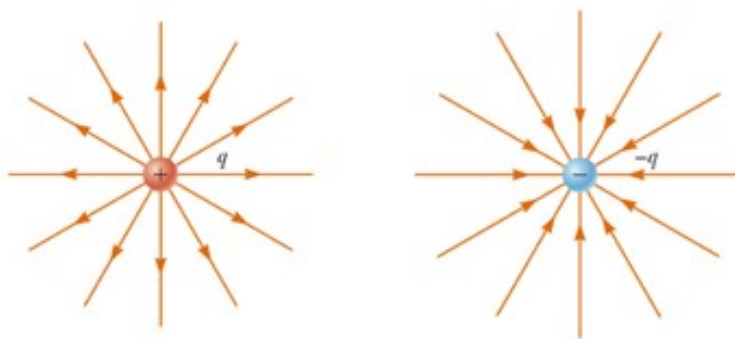


Figure 1: Electric Field Lines for isolated positive and negative point charges.

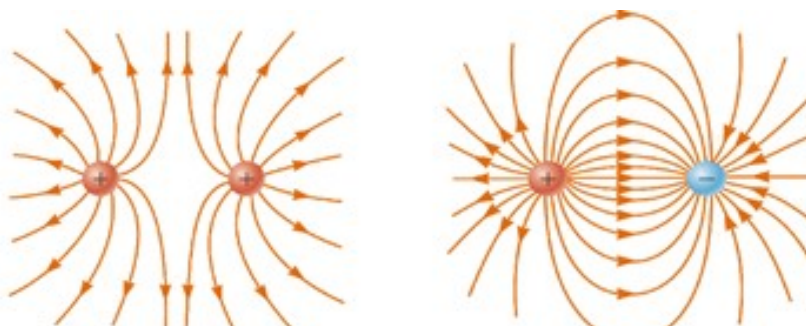


Figure 2: Electric Field Lines for two similar and dissimilar charges.



Figure 3: Electric Field lines (Orange) and Equipotential Lines (blue) for a uniform electric field, point charge and two opposite charges.

34.4 PROCEDURE

Preparing the Conducting paper

1. Use the Conductive Ink Pen to draw the desired shapes on the conductive paper. You may use the stencil to draw neat circles etc.
2. This should be done in an open area as the ink releases harmful fumes.
3. Let the ink dry completely before touching the drawing or stacking the papers on each other.

Equipment Setup

1. Place the conducting paper with electrodes painted on them on the corkboard of the field mapping board. Attach with pins to keep it in place.
2. Join and tape the two prongs of the multimeter together so that the distance between the probes is fixed at about 1 to 2 cm.
3. Connect the positive and negative terminals of the power supply (max 15 V dc), one to each electrode. Use the conducting pins to do this.
4. To use the 850 Interface as a power supply, click on Signal Generator. Set Output 1 to 15 V DC signal. Click Auto, so that the power will come on when you press Record. Clicking on Signal Generator a second time will close it.
5. You may also use an external DC supply.

6. First test the electrodes by measuring the voltage at different points on the surface of the electrode. If it is not the same everywhere on the electrode, there may be a break in the painted electrode.

Electric Field Lines

7. Place the negative probe at a point close to the negative electrode. Call this 'point 1'. Using this as a pivot, move the other electrode around to find the position where the voltage difference is maximum. Call this location 'point 2'.
8. Record the two points on the graph paper that has the same pattern drawn on it as the conducting paper.
9. Pick up the negative probe from 'point 1', and place it at 'point 2', and move the positive probe around until you find the maximum voltage difference. Call this 'point 3' and record it on the graph paper. Continue this way until you reach the positive electrode.
10. Now join all the points with a smooth line. This is now your first electric field line. Mark arrows to show the direction of the electric field line (from positive towards negative).
11. Now select another point close to the negative electrode and draw another electric field line. Repeat this until you have enough lines to clearly show the pattern of the electric field. Make sure there are no big gaps between the field lines on your paper.

Equipotential Lines

12. Place the negative probe near the center of the field. Rotate the positive probe until you find the point where the potential difference is zero. Note these points on the graph paper.
13. Move the negative probe to where the positive probe found the zero reading, and repeat, until you reach back to the starting point, or the edge of the conducting paper. Join the dots to obtain an equipotential line. (use a different color or line type than the one you used for the electric field lines)
14. Repeat for several other starting points until you have a nice view of the equipotential lines.
15. Label the lines and charges.
16. Your final graph for two, point charges should look something like that shown in Figure 4.
17. Make the electric field lines and equipotential lines for other configurations of the electrodes.

Equipotential Contour Map

18. Use the sheet with the two electrodes.
19. Connect one electrode to the negative and the other to the positive voltage.
20. Place the negative probe of the voltmeter at the negative electrode.
21. Place the other probe of the voltmeter at each of the lines of intersection on the grid of the conducting paper and note the voltage.
22. Write this voltage on an excel sheet, with the data along the X-axis along the rows, and the data along the Y-axis along the columns.
23. In excel, select the whole data, and select: INSERT – CHARTS – ALL CHARTS – SURFACE.
24. This will give you a 3-dimensional chart with the X and Y axes along the sides of the paper, and the Z-axis as the voltage. It is color coded to show the contours of equal voltage.
25. You can rotate this by 90 degrees by selecting: DESIGN – SWITCH ROW/COLUMN

34.5 CALCULATIONS

There are no calculations in this Lab.

34.6 PRECAUTIONS

1. Make sure that the power supply is well connected to the electrodes on the conducting paper.
2. When you touch the conducting paper with the pointed voltmeter prongs, take care not to puncture the paper.
3. The image of the conducting paper and the electrodes should be the same on the graph paper on which you are drawing the lines.
4. Do not have large regions on the paper where there are no lines.
5. Draw arrows in the Electric Field lines, but not on the Equipotential lines. The direction of the arrows in the EF lines go from positive to negative terminals.
6. Label the two types of lines clearly. You may use different colored pencils to do this.

34.7 DATA

There are no Data Tables in this Lab.

Make the Electric Field Lines and Equipotential Lines for the following configurations (or as directed by your instructor):

1. Single point charge (positive or negative)
2. Two similar point charges
3. Two dis-similar point charges
4. Two parallel plates (will look like lines on the conducting paper)

34.8 REPORT

Upload the following in the DataSet for this Lab.

		Points
1.	1 Electric Field and Equipotential lines drawing for Case 1	15
2.	5 3 Electric Field and Equipotential lines drawing for Case 2	15
3.	4 Equipotential Contour Map Data Table	10
4.	5 Equipotential Contour Map	10
5.	6 Sources of Error in this experiment. Indicate the major source of error.	5
6.	7 Discussion of Results	10
	Total	65

34.9 ADDITIONAL INFORMATION

For practice, and getting a good idea of the Electric Field and Equipotential lines for different configurations, you can draw them by using the simulation:

https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html

34.10 POINTS TO THINK ABOUT

1. Why should Electric Field lines have arrows, but not on the Equipotential lines?
2. Why are the Electric Field Lines and Equipotential lines perpendicular to each other where they intersect?
3. Do any of the Electric Field lines intersect each other? If not, why?
4. Do any of the Equipotential lines intersect each other? If not, why?
5. If you made a circle (ring) as the electrode on the conducting paper, what would the EF and EP lines look like inside the circle?
6. In the previous case, what is the electrode were a rectangle instead of a circle?

34.11 SAMPLE DATA

Electric Field Lines and Equi-Potential Lines

An example of the Electric Field Lines and Equi-Potential lines is shown in figure 4.

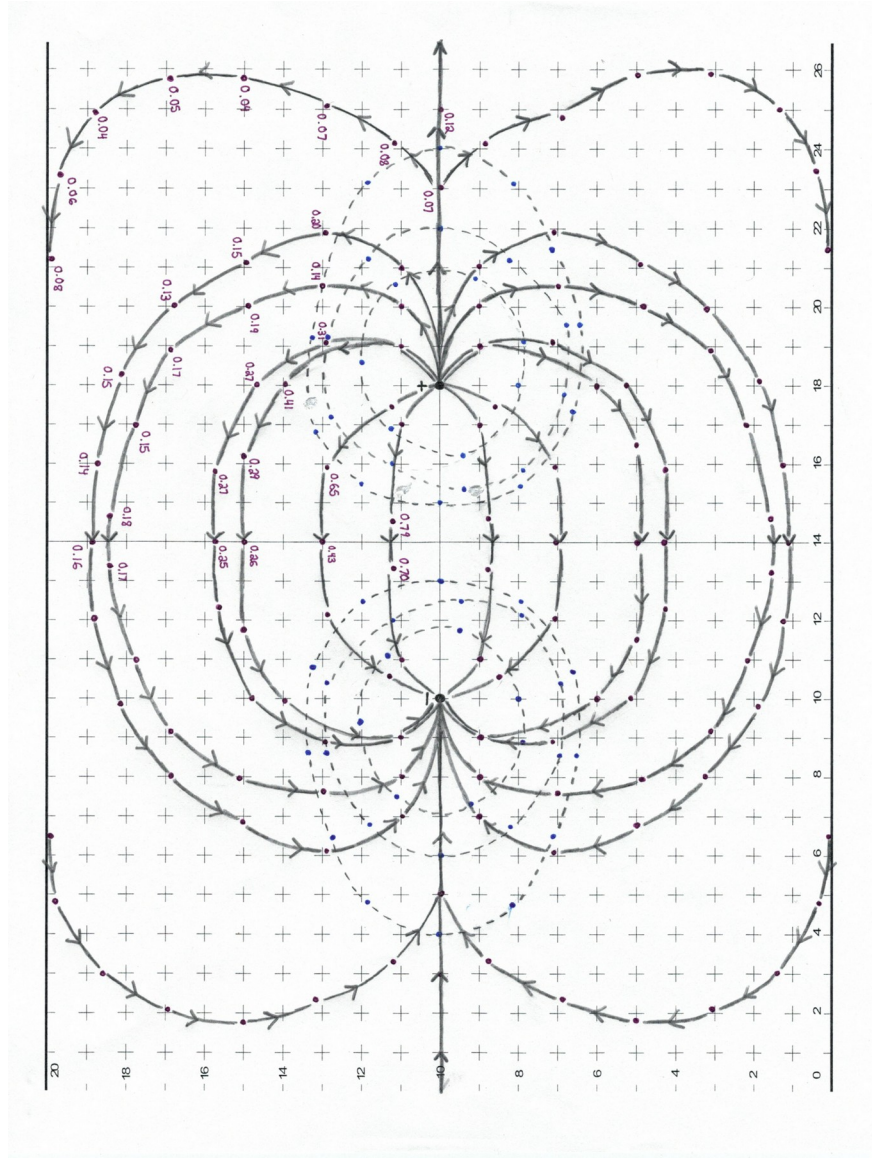


Figure 4: Electric Field and Equipotential Lines for two point-charges. (Figure taken from an earlier semester student report).

Equipotential Contour Map

An example of this is shown here with some random data. Here the data is both positive and negative. Your map will be all positive values, but the shape would be similar. If you like, you can subtract $V/2$ from the entire data, and get a contour map with + and - values.

Random data and contour map for equipotential lines.

			1	1	1	1	1	1	1	1				
			1	-1	-2	-2	-1	-1	-1	1				
			1	-1	-3	-3	-4	-4	-2	1				
			1	0	-1	-1	-2	-2	0	1				
			1	0	0	0	-1	-1	0	1				
			1	1	1	1	1	1	1	1				
			1	1	2	3	3	3	2	1				
			1	1	2	5	5	4	3	1				
			1	1	1	3	3	2	1	1				
			1	1	1	1	1	1	1	1				

